Predicting Energy Consumption Based on Building Attributes

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With the advent of generative AI, there has been a need to build data centers across the United States. However, these data centers have become extremely controversial in the communities they are built in, as they can put a massive strain on the power grids of these communities, even causing blackouts at times.[1] This is just one of numerous examples of why the power grid in the United States needs to be updated. Even the U.S. Department of Energy agrees that something needs to be done.[2] However, these changes can't be done overnight and will require a few years of planning and implementation before the grid can be fully updated. Therefore, something needs to be done in the meantime. This project aims to tackle this challenge. By building a model that can predict the energy usage of a building, the energy efficiency of the power grid. This can help identify certain "problem" buildings that are way above their expected energy consumption; along with this, building managers can use these predictions themselves to better optimize their energy consumption. The model was trained and tested using the Energy and Water Data Disclosure for Local Law 84(LL84) dataset from 2022, which is provided by New York City's Mayor's Office of Climate and Environmental Justice.

The LL84 dataset is a very extensive dataset. In total, it has data for nearly 30 thousand buildings. It also has 249 columns in total. Most of these can be easily cut from being used since a good portion of these columns are either non-numeric or are for a specific type of building, such as the Bank Branch – Gross Floor Area (ft^2) column, which only has data for a small percentage of the buildings, making it not useful for our model. Source EUI was the dependent variable selected. This was a measurement of the total energy used, measured in kilo British

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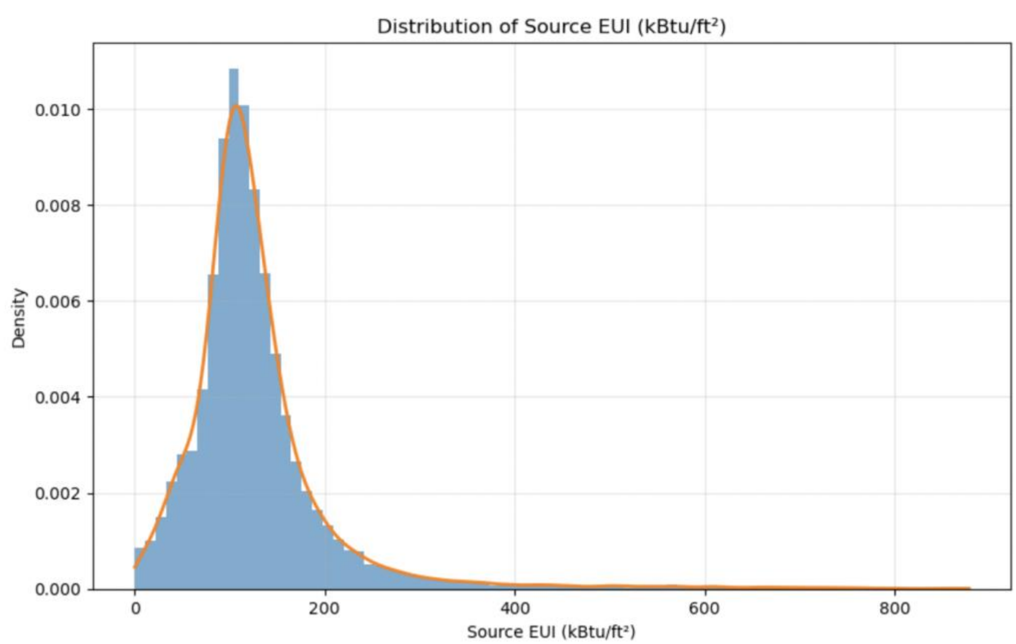
[1] *CNBC*, "Texas Data-Center AI Boom Puts Pressure on ERCOT, Raising Blackout Risk," November 22, 2025, https://www.cnbc.com/2025/11/22/texas-data-center-ai-ercot-blackout-power-outage.html

[2] U.S. Department of Energy, "Department of Energy Releases Report on Evaluating U.S. Grid Reliability and Security," July 7, 2025, https://www.energy.gov/articles/department-energy-releases-report-evaluating-us-grid-reliability-and-security

thermal units, which is then divided by the total square footage of the property (kBtu/ft^2). Some notable columns in the dataset include the year the building was built, the ENERGY STAR rating of the property, electricity use, greenhouse gas emission intensity, number of bedrooms, and number of residential living units. It's important to note that an ENERGY STAR rating rates the energy efficiency of a property on a scale of 0 to 100, with 100 being the most efficient. Active IT meters refer to energy meters that are being tracked but do not contribute to the property's energy use. This may be something like a parking meter or a submeter.

The model selected for the project was a Gamma regression. This choice was motivated primarily by the characteristics of the distribution of the response variable, Source EUI (kBtu/ft^2). The data exhibits substantial right skewness, which is a defining feature of the Gamma family, and all values were strictly positive, which is an essential requirement for Gamma models. All of these characteristics can be seen in the density plot below.



Additionally, the spread of the data increased as the mean increased, suggesting non-constant variance. This pattern is consistent with the Gamma variance function, where, $\text{Var}(Y_i) = \phi\mu_i^2$, meaning the variance grows quadratically with the mean through a constant dispersion parameter ϕ, this can be thought of as like a scale for the distribution. These properties make the Gamma distribution appealing in situations where linear regression, which assumes normally distributed errors and constant variance, may be inappropriate.

For this project, a log-linked Gamma regression was used, which is the most common version of the model. The log link function applies the natural logarithm to the expected value of the response, transforming the model so that the log of the mean is a linear function of the predictors. This transformation ensures that all predicted values are positive and also turns multiplicative effects in the original scale into additive effects on the log scale.

The workflow of the project can be broken down into a few parts, the first of which was cleaning and preparing the dataset. This initially was very minimal on the first attempt of running the gamma regression; however, this would become very extensive after numerous attempts. The first preparation dealt with outliers in our target variable. Gamma regression is extremely affected by outliers, leading to the need for them to be cut from the dataset. Another thing that came up was that the LL84 dataset is formatted weirdly. For example, if an element in a column had no entry, instead of putting NaN, a string was put in that said, “Not Available”. This proved to be a problem, as when the data was stripped down to just columns that contained numerical variables, columns such as the ENERGY STAR rating were removed from the dataset, even though it’s a rating from 0 to 100. This meant that the object columns had to be overridden to be numeric columns. Several of the columns in the LL84 dataset contained extreme outliers. Due to this, the columns were clipped of their bottom and top 1% values to avoid outliers. After this, the data was split into a training set, a validation set, and a testing set. 70% of the data was placed in the training set, while the other 30% was split in half, with 15% going to the validation set, and the last 15% going to the testing set. Once this was done, StandardScaler was used to standardize all three sets of data. At this time, the model's MSE was still terrible, and more needed to be done to prepare the data for fitting. Therefore, since the model appeared to have a good amount of collinearity in its predictors, the data was put through VIF testing. A function was made to compute the VIF of each predictor. The data was then looped through with this function, in which all predictors with a VIF greater than 10 were dropped from the dataset.

After running the model a few times, it became apparent that there were some problematic predictors making their way into the model. One of these was "Property Id," which is just the unique identifier for each building; therefore, it has no actual meaning and needed to be removed. Another extremely troubling predictor was "Number of Active IT Meters", as it had a coefficient of 0 and had no variation, meaning it was constant throughout the dataset; therefore, it was also removed. The code for this process had to be moved earlier in the process, in order for the code in later cells to work properly. Finally, one last bit of preparation that needed to be done was to quote the column names. Most of the columns in the LL84 dataset contained spaces and parentheses, which statsmodels can't handle. A solution was to make a quote for each column; that way, it wouldn't try to interpret any symbols it found in the name of the columns. After each column was properly quoted, the regression formula was built with the quoted columns so that the model could finally be fitted. The formula, along with the training data frame, was used to build the model.

Backward stepwise selection was used to put the most meaningful predictors into the Gamma regression model. Backward stepwise selection takes the full model and iteratively removes the least useful predictor until the remaining variables are all statistically significant. To determine how statistically significant a predictor was, a p-value test was used. A p-value represents how close a predictor is to being statistically significant, with a smaller p-value representing a higher chance. In the code, the p-values for all the predictors were gathered. Predictors are then removed iteratively, beginning with the variable exhibiting the highest p-value, the process continues until all remaining predictors achieve statistical significance at the 0.05 level. This is the normal threshold used to say that a predictor is statistically significant. During the loop, the model has to be refitted after the removal of every predictor, as the removal of a variable can affect the p-values of other predictors. After the loop is done, the final set of predictors is extracted so that they can be used for our ordinary least squares regression later on.

The model is then used with the validation set data, to see how well it works with new, unseen data. The current model that was used on the training set is taken and used to generate the predicted values for the validation set. In order to see how well the model is doing the Mean Squared Error (MSE) and Mean Absolute Error (MAE) are taken. The MSE takes the sum of the squared residuals and divides it by the total of entries to get the average of the squared error. The MAE takes the sum of the absolute value of the residuals and divide it by the total number of entries to get the average. The MSE for the validation set 4994.11 while the MAE was 27.62. This means that on average, the model's predictions were 27.62 kBtu/ft^2 away from the actual value. The square root of the MSE can be taken to get the Root Mean Square Error (RMSE), which will tell of the average magnitude of the errors. The square root of the MSE is about 40.67. Both of these values show that the model is having a decent performance and is properly fitted.

After selecting the final set of predictors using backward stepwise selection and confirming model performance on the validation set, the final Gamma regression model was retrained using the combined training and validation data. This step increases the effective sample size for estimating the regression coefficients, thereby reducing estimator variance and improving numerical stability. This will help improve the performance of the model by increasing the precision of the parameter estimates.

Once the model has been retrained, the test set is used as an unbiased estimate of how well the model is expected to perform on new buildings. The predictions from the model are gathered, and once again, the MSE and MAE are used to see how well the model performed. The MAE for the model was 27.23, which means that, on average, the model is about 8.39 kBtu/ft^2 away from the actual value. This lines up well with our MAE of 27.62 from the validation set and shows that the model is not overfitting. The MSE is 4839.40, which converts to an RMSE of 69.57. This is roughly similar to the MSE from the validation set, considering the data, these results are not too bad.

Next, the prediction intervals for the model were computed. As previously discussed, the variance of the Gamma GLM is a function where the variance is equal to the mean of the point squared multiplied by a dispersion parameter, which is called scale in statsmodels. This is why the scale must be set before the prediction intervals are calculated. 500 simulations were done to calculate the intervals, with each simulation being put in a matrix where each row represents a building and the columns represent a simulation result. In the loop of the simulations, the shape and scale parameters are calculated, where the shape is equal to 1 over the calculated scale from before, and the scale parameter is equal to the calculated scale times the mean. These parameters are used by statsmodels to simulate the predicted outcome. This process is repeated until the 500 simulations are completed.

After filling up the matrix with the simulation results, the lower and upper bounds of the intervals are calculated, where the lower bound is the 2.5th percentile and the upper bound is the 97.5th percentile. These intervals show where a future observation is expected to be when it has the same conditions as the calculated interval. The summary statistics for the prediction intervals were also calculated. The mean interval width and the standard deviation of interval width were used. Mean interval width adds up the widths of every interval and divides them by the number of intervals to get the mean. The standard deviation of interval width takes the sum of the squared difference of the width and the average width, which is all divided by the number of intervals minus one. The square root of that is then taken to get the standard deviation. The mean prediction interval width was about 137.95 kBtu/ft^2. This shows that there is a good amount of uncertainty present in the model.The standard deviation of interval widths was about 128.36. This is relatively high compared to the mean width, which shows that the intervals for the buildings can vary widely. This can be expected, though, and is not a huge concern for the model.

Next, an ordinary least squares regression model was built to compare how it performs versus the Gamma GLM. The predictors from the Gamma regression are taken and stripped, for the Q("…") format, as the actual column name is needed. Next, the predictor matrices and target vectors are built. A constant column is also added, which will be used to put the intercept in so that it is present in the model. The indices of the matrix and the target vector are reset to ensure that they are matched up perfectly. The training set response vector and the predictor matrix are then fitted to the OLS model. The coefficients of the model are calculated by the inverse of the transposition of the predictor matrix multiplied by the predictor matrix. This is then multiplied using matrix multiplication by the transposition of the predictor matrix. Finally, the result of that computation is multiplied by the response vector using matrix multiplication again. This gives us the fitted OLS model.

Just like with the Gamm regression, the MSE and MAE of the validation set are computed. The MSE is 1552.99 kBtu/ft^2 while the MAE is 24.50. This means on average the OLS model is less than 25 kBtu/ft^2 away from the actual value. The RMSE of the model is about 39.41. This shows that the model is performing extremely well before retraining and is also doing much better than the Gamma GLM. After this, the training and validation sets were combined, and the design matrices were constructed again, along with a new intercept term being added. Finally, the indices were reset, and the model was fitted again. This final model was used on the test set, giving us an MSE of 1361.45, which equals an RMSE of 36.90 and a MAE of 24.16. The model is performing significantly better than the Gamma model, even though the data follows a Gamma distribution.

Lastly, the prediction intervals for the OLS model were calculated and put into a summary frame. The lower and upper bounds for an alpha level were extracted, and the interval widths were calculated by subtracting the lower bound from the upper bound. The mean and standard deviation of the widths of the prediction intervals were calculated. The mean width was 143.36, which was slightly worse than the Gamma regression model. The standard deviation was only .05. This is significantly smaller than the standard deviation of the Gamma model. This means that the model assigns almost identical uncertainty for all buildings. This is the exact opposite of what was seen in the Gamma model, where there was a high degree of variation in the widths.

Despite the fact that the data followed a gamma distribution, and not a normal distribution, the OLS model performed better than the Gamma model. Looking at the MSE of the Gamma model, the validation was 4994, while the test was 4839. Meanwhile, the OLS model had an MSE of 1353 and 1361. These are vastly better and much more expected results. The reason for the better performance could be caused by the Gamma model penalizing large errors much more than the OLS. Along with this, the variance may not follow the Gamma variance function as much as initially anticipated. There was also a significant number of predictors lost due to the cleaning. Despite this, both models performed decently well with the prediction intervals. The Gamma model had a mean prediction interval width of 138, while the OLS had a width of 143. However, there was much more variation in the Gamma's intervals compared to the OLS's intervals. Nonetheless, both of these models performed extremely well with the energy data. They were able to show that a relatively small set of interpretable, physical predictors can explain a good portion of the variation in building energy intensity.

Looking into the future, there can be multiple ways that can be used to improve the models further. Looking at interaction effects can be beneficial, as these types of predictors were not used at all in the model. Another thing to look at is looking at using LASSO instead of backward stepwise selection. Backward stepwise selection is known to be sensitive to noise in the data. Along with this, using LASSO can help reduce the overfitting that seemed to be present in the Gamma model. Looking at other regressions could be an alternative step. For example, robust regressions, such as quantile, are known to handle outliers extremely well, unlike with Gamma regression, which is extremely sensitive to outliers which may have affected its performance. Another type of Gamma regression could also be used, such as lognormal or tweedie. Building more models and comparing them to our two already established models can help find what problems are affecting the Gamma model the most, which can be addressed and thus improve the initial Gamma model. This project has been a good start into building a model that can eventually help out the U.S. power grid, which is in need of some sort of relief at this moment.

References

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2. U.S. Department of Energy. 2025. “Department of Energy Releases Report on Evaluating U.S. Grid Reliability and Security.” July 7, 2025. https://www.energy.gov/articles/department-energy-releases-report-evaluating-us-grid-reliability-and-security